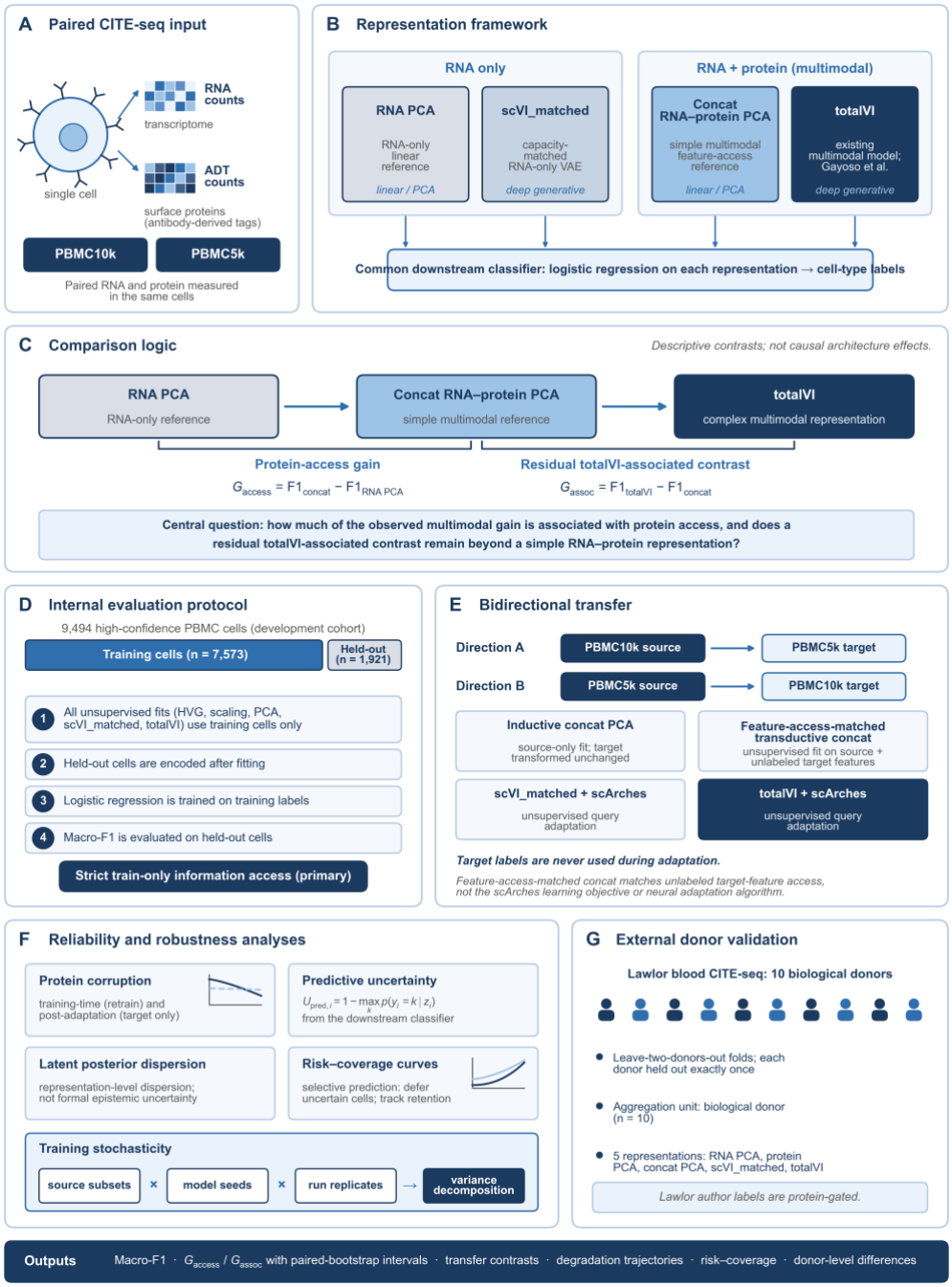
Supplementary Information

Evaluating protein-access gains and residual totalVI-associated contrasts in CITE-seq representation learning under distribution shift

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Contents: Supplementary Figure S1; Supplementary Methods S1–S10; Supplementary Results SR1–SR6 with Supplementary Tables S1–S14; Supplementary Data (05_GPB_Supplementary_Data.zip) file index. Primary internal results use the strict train-only protocol (7,573 training / 1,921 held-out high-confidence cells). Bootstrap intervals are cell-level conditional intervals, not biological confidence intervals. Seeds, folds, and run replicates are computational units; the only biological replication unit is the Lawlor donor (n = 10).

Supplementary Figure S1



Supplementary Figure S1 | Overall study and reliability-evaluation design. (A) Paired CITE-seq input from the public PBMC10k and PBMC5k datasets. (B) Representation categories: RNA-only (RNA PCA, scVI_matched) and multimodal (concatenated RNA–protein PCA, totalVI), all scored with the same downstream logistic-regression classifier. (C) Comparison logic defining the protein-access gain (G_{access}) and the residual totalVI-associated contrast (G_{assoc}), both descriptive contrasts rather than causal architecture effects. (D) Primary strict train-only internal evaluation (7,573 training / 1,921 held-out high-confidence cells); the historical feature-transductive fit is a secondary sensitivity analysis. (E) Bidirectional transfer (Direction A, PBMC10k→PBMC5k; Direction B, PBMC5k→PBMC10k) comparing source-only inductive concat, the feature-access-matched transductive concat control, and scArches query adaptation of scVI_matched and totalVI; target labels were never used during adaptation. (F) Reliability and robustness analyses: training-time and post-adaptation protein corruption, classifier predictive uncertainty, latent posterior dispersion, risk–coverage and selective prediction, and training stochasticity with variance decomposition. (G) Lawlor donor-level external validation (10 biological donors; author labels are protein-gated). Representation construction is detailed in main Figure 1; quantitative results are shown in main Figures 2–6. The schematic contains no experimental results; small curve icons in panel F are illustrative glyphs, not data.

Supplementary Methods

All parameters below were verified against repository code, configuration files, logs, or frozen outputs.

S1. Dataset provenance and quality control

Table S1 | Internal PBMC CITE-seq data.

Item	Verified value
Source files	pbmc_10k_protein_v3.h5ad, pbmc_5k_protein_v3.h5ad (scvi-tools exampledata; YosefLab/scVI-data fallback)
Loader	Project loader (src/data/load_citeseq.py); scvi.data.pbmcs_10x_cite_seq is not the primary API
Combined object	Inner join: 10,849 cells × 15,792 genes × 14 proteins
Per-dataset cells	PBMC10k 6,855; PBMC5k 3,994
Additional cell/gene QC in project	None (additional_cell_filter: false); inherited from processed source objects
GEO/SRA accession	Not claimed (not present in local metadata)

Table S2 | Lawlor external cohort.

Item	Verified value	Source
HCA / ENA	efea6426-510a-4b60-9a19-277e52bfa815; PRJEB40376 / ERP124005	Project metadata
Singlets	16,175	Data-preparation summary
Author-labeled cells	5,207	Data-preparation summary
Harmonized genes	12,776	Data-preparation summary
Matched proteins	12	Data-preparation summary
Biological donors	10	Fold and donor tables
Gene mapping	Ensembl → symbol; 55 duplicated symbols aggregated by summing	Gene report
Protein aliases	Alias-harmonized to internal names (e.g., PD-1 ↔ CD279)	Protein overlap table
Labels	Author labels; protein-gated	Main-text Methods
Batch key for deep models	Sequencing lane / run_identifier (donor not used as batch)	Data-preparation summary

S2. PCA and concatenated-PCA preprocessing

Table S3 | Linear representation preprocessing and fit scope.

Step	Setting	Fit scope (development)	Fit scope (inductive transfer)
Highly variable genes	2,000; seurat_v3	Training HC cells	Source
RNA normalization	Total count 1e4 → log1p → scale, clip ±10	Training	Source
RNA PCA	10 PCs	Training	Source
Protein	CLR (pseudocount 1) → z-score (ddof=1) → 10 PCs	Training	Source
Concat	10 + 10 PCs → StandardScaler	Training	Source
Transductive concat control	Same unsupervised steps on source + unlabeled target	—	Joint unlabeled features; classifier source-only

Note: config.yaml lists n_top_genes: 4000 for historical settings; the primary PCA path hard-codes 2,000.

S3. Deep model hyperparameters and information access

scVI_matched (primary RNA-only deep comparator). n_latent=20, n_hidden=256, n_layers=2, dropout_rate=0.2, gene_likelihood="nb", dispersion="gene", latent_distribution="normal", use_observed_lib_size=True; max_epochs=200, early stopping patience 45, train_size=0.9, batch_size=256. Learning rate: scvi-tools TrainingPlan default (not overridden).

scVI default (not the primary reported comparator). n_hidden=128, n_layers=1, gene_likelihood="zinb", dropout_rate=0.1; retained only for historical PHASE-3 runs.

totalVI. n_latent=20, n_hidden=256, encoder layers 2 / decoder layers 1, dropout 0.2, gene_likelihood="nb", protein_dispersion="protein"; lr=4e-3, reduce_lr_on_plateau=True; same epoch, early-stopping, and batch settings.

Internal deep-model information access — PRIMARY (strict train-only; matched information access). PCA, scVI_matched, and totalVI unsupervised fits used only the 7,573 high-confidence training

cells. The 1,921 held-out high-confidence cells were encoded after fitting (frozen PCA transform; scVI/totalVI get_latent_representation with transferred AnnData setup; no query adaptation). Classifier: logistic regression on training labels only; scoring on held-out cells only. Hyperparameters unchanged (n_latent=20, matched capacity, max_epochs=200, patience 45, batch_size=256, totalVI lr=4e-3). Status: COMPLETE — PRIMARY ANALYSIS.

Supplementary sensitivity (historical feature-transductive). Earlier repository deep models were trained on the full combined object (10,849 cells); obs['split'] was not used for deep-model subsetting, and train_size=0.9 is an internal ELBO validation split only. That unequal-access evaluation yielded RNA PCA 0.666, concat 0.745, scVI_matched 0.722, and totalVI 0.779, with $G_{assoc} \approx +0.034$ [−0.012, 0.073]. It is not the primary internal result (see SR1b).

scArches transfer. Source training → prepare/load query → unsupervised adaptation (query_max_epochs=200, weight_decay=0) → encode both cohorts with the adapted model → logistic regression on post-adaptation source coordinates → target scoring. Target labels were withheld until scoring. The feature-access-matched concat control matches unlabeled target feature access but not the objective, architecture, optimization, or adaptation algorithm.

S4. Classifier

sklearn.linear_model.LogisticRegression with max_iter=2000, solver="lbfgs", random_state=seed; defaults C=1.0 and class_weight=None. The same strategy was used for every representation.

S5. Internal split chronology

1. Stratify by batch, test_fraction=0.20, seed 0, on 10,849 cells → exactly 8,679 train / 2,170 test.
2. Restrict to high-confidence labels → 7,573 / 1,921 (9,494 high-confidence cells).
3. PRIMARY: unsupervised fits for PCA, scVI_matched, and totalVI on high-confidence training cells only; held-out cells encoded after fitting.
4. Classifier trained on training labels only; evaluated on held-out high-confidence cells.
5. Supplementary sensitivity: historical deep models fit on all 10,849 cells' features (feature-transductive).

Label audit: RNA-only re-annotation agreement 1.0 on the 9,494 high-confidence cells.

S6. Bootstrap

Primary: cell-level global paired bootstrap, B = 10,000, seed 20260923, percentile 2.5/97.5. Sensitivity: class-stratified paired bootstrap (within-class resampling), seed 20260924, same B. Intervals are conditional on fixed prediction vectors; they are not biological confidence intervals and not donor-level intervals.

S7. Degradation designs

Entry-wise masking of originally nonzero protein counts (preexisting zeros unchanged) at fractions 0.00, 0.10, 0.25, 0.40, 0.55, 0.70, and 0.85. Training-time (historical feature-transductive secondary analysis): totalVI retrained; five mask/retraining replicates at nonzero levels; model_seed=0; clean n = 1. Post-adaptation: target proteins corrupted only after clean adaptation; classifier frozen. Marker-channel dropout (SR5): entire protein channels zeroed before protein/concat PCA; totalVI not retrained.

S8. Uncertainty and selective prediction

Predictive uncertainty $U_{pred,i} = 1 - \max_k p_k$ from the downstream logistic regression. Latent dispersion = mean posterior variance over latent dimensions (not epistemic uncertainty). Primary selective risk uses accuracy; macro-F1 and balanced accuracy versus coverage are sensitivities (SR3).

S9. Stochasticity design and variance model

Balanced design: 5 source subsets × 5 model seeds × 2 run replicates. For representation a, $Y_{ijr}^{(a)} = \mu^{(a)} + S_i^{(a)} + G_j^{(a)} + (SG)_{ij}^{(a)} + \varepsilon_{ijr}^{(a)}$, and paired differences $\Delta_{ijr} = Y_{ijr}^{(totalVI)} - Y_{ijr}^{(scVI_matched)}$. Source, seed,

interaction, and residual are computational components, not biological variance. Variance fractions used method-of-moments expected mean squares.

S10. Software environment

Table S4 | Software versions (verified project environment logs / pip show).

Package	Version
Python	3.11.7
scvi-tools	1.3.3
scikit-learn	1.5.2
scanpy	1.10.3
anndata	0.11.4
PyTorch	2.14.0
numpy	1.26.4
pandas	2.2.2
scipy	1.13.1
mofapy2	0.7.5
CUDA	Local Mac logs: not available (MPS available); PHASE 10 server runs used CUDA builds

Pins: environment.yml and requirements.txt in the code repository (https://github.com/Jizhuy/citeseq-multimodal-reliability).

Supplementary Results

SR1. Class-stratified paired bootstrap (sensitivity)

Primary intervals remain the global cell-level paired bootstrap under the strict train-only protocol (B = 10,000, seed 20260923).

Table S5 | Global versus class-stratified paired bootstrap (strict train-only).

Quantity	Global 95% interval (primary)	Class-stratified 95% interval	prop>0 (stratified)
G_{access}	[0.042, 0.131]	[0.046, 0.120]	1.000
G_{assoc}	[−0.060, 0.030]	[−0.035, 0.023]	0.358
Gap_{total}	[0.029, 0.130]	[0.030, 0.123]	1.000
R_{access}	[0.680, 2.304]	[0.768, 1.968]	—

Interpretation: stratification narrows the intervals but does not change the conclusions; the protein-access contrast remains entirely above zero, and the residual totalVI-associated contrast remains near null and still crosses zero. Data: strict_train_only/strict_train_only_bootstrap_global_vs_stratified.csv.

SR1b. Effect of held-out feature access on the internal deep-model comparison (not primary)

Table S6 | Historical feature-transductive versus strict train-only (PRIMARY) internal results.

Metric	Feature-transductive (historical)	Strict train-only (PRIMARY)	Notes
RNA PCA	0.666	0.667	PCA was train-only in both
concat PCA	0.745	0.745	PCA was train-only in both
scVI_matched	0.722	0.709	Deep-model access changed
totalVI	0.779	0.741	Deep-model access changed
G_{access}	+0.078 [0.043, 0.132]	+0.078 [0.042, 0.131]	Essentially unchanged
G_{assoc}	+0.034 [−0.012, 0.073]	−0.004 [−0.060, 0.030]	Attenuated
Gap_{total}	+0.112 [0.084, 0.154]	+0.074 [0.029, 0.130]	Reduced with train-only deep models
R_{access}	≈0.70 [0.405, 1.114]	≈1.05 [0.680, 2.304]	Secondary descriptive ratio only

Historical point values are taken from the frozen historical bootstrap output (G_{access} 0.0784, Gap_{total} 0.1124). Earlier summaries listed +0.079 and +0.113, which are differences of rounded F1 values (0.745 − 0.666; 0.779 − 0.666).

In the historical analysis, scVI_matched and totalVI were fit using features from all 10,849 cells while PCA remained train-only. Under the primary strict train-only protocol, all representations were fit on the same 7,573 high-confidence training cells and evaluated on the same 1,921 held-out cells. The apparent totalVI advantage over concat was substantially attenuated when held-out-cell feature access was removed. Deep-model performance, particularly totalVI, was sensitive to the representation-learning information-access regime. This does not claim that transductive access alone causally explains the entire numerical difference; it documents that the information-access regime can materially affect apparent representation-level gains.

SR2. Cell-type-specific post-adaptation degradation (Direction B, totalVI)

Table S7 | Mean per-class F1 (over seeds) at clean versus protein corruption 0.85.

Cell type	F1 clean	F1 at 0.85	ΔF1
CD8 Naive	0.784	0.519	−0.264
CD8 TCM	0.261	0.189	−0.072
CD8 TEM	0.826	0.755	−0.071
CD4 Naive	0.853	0.797	−0.056
CD4 TCM	0.712	0.668	−0.043
CD4 TEM	0.575	0.533	−0.042

CD8 Naive shows the largest drop among the focus subsets (≈ −0.26). Descriptive only; seed means are computational, not biological replicates. Data: degradation/class_degradation_dirB_focus_subsets.csv (all classes: degradation/class_degradation_dirB_totalVI_p085_delta.csv).

SR3. Selective prediction: accuracy versus macro-F1 and balanced accuracy

Table S8 | Direction B totalVI at 50% versus 100% coverage (mean over 20 seeds).

Coverage	Mean accuracy	Mean macro-F1	Mean balanced accuracy
1.00	0.787	0.654	0.659
0.50	0.966	0.703	0.698

Primary selective risk uses accuracy. Retention remains cell-type imbalanced (main text). Data: selective_prediction/selective_prediction_dirB_totalVI_cov50_vs_100.csv and selective_prediction/selective_prediction_macroF1_by_coverage.csv.

SR4. Lawlor donor-level summaries

Table S9 | Paired totalVI − scVI_matched present-class macro-F1 (seed-averaged within donor; n = 10 donors).

Statistic	Value
Mean paired Δ	+0.090
Median paired Δ	+0.089
SD	0.010
IQR	0.013
Min / Max	+0.077 / +0.110
Donors favoring totalVI	10/10
Exact two-sided sign test P	0.00195

Table S10 | Donor-mean present-class macro-F1 for all five representations (n = 10 donors).

Representation	Mean	Median	SD	IQR	Min	Max
RNA PCA	0.690	0.698	0.031	0.016	0.626	0.730
Protein PCA	0.923	0.920	0.013	0.017	0.898	0.940
Concat PCA	0.942	0.943	0.010	0.013	0.927	0.954
scVI_matched	0.794	0.787	0.016	0.030	0.772	0.815
totalVI	0.884	0.883	0.013	0.017	0.857	0.897

Data: lawlor/lawlor_all_methods_donor_level_summary.csv, lawlor/lawlor_all_methods_per_donor.csv, lawlor/lawlor_donor_summary_stats.csv, lawlor/lawlor_donor_paired_deltas_present_class.csv.

Historical fold-mean PCA aggregates (continuity only; not the preferred aggregation for main Figure 6): protein PCA ≈ 0.927 and concat ≈ 0.945 (lawlor/lawlor_simple_baseline_fold_means.csv).

Latent consistency favoring totalVI was 38/50 matched fold × seed comparisons. These are computational comparisons, not biological replicates; the biological n is 10 donors.

Fixed-ontology Lawlor sensitivity: the audited seed-mean tables include a fixed-ontology macro-F1 column that is identical to present-class macro-F1 for these donors; no separate divergent sensitivity analysis was run (status PARTIAL in Table S13).

SR5. Marker-channel dropout (PCA baselines only; totalVI not retrained)

Table S11 | Development-cohort macro-F1 after zeroing protein channels.

Setting	Protein PCA	Concat	Δ concat vs clean
Clean	0.512	0.745	0.000
Drop CD3	0.457	0.770	+0.025
Drop CD4	0.448	0.762	+0.017
Drop CD8a	0.485	0.755	+0.010
Drop CD14	0.469	0.742	−0.003
Drop CD19	0.463	0.754	+0.009
Drop CD56	0.465	0.738	−0.007
Panel T-lineage (5 markers)	0.328	0.720	−0.025
Panel myeloid (3 markers)	0.421	0.750	+0.005
Panel B/NK (2 markers)	0.467	0.741	−0.004

Clean concat macro-F1 matched the frozen baseline (0.745). Exploratory PCA-only sensitivity; it does not test totalVI under panel failure. Data: marker_channel_dropout/marker_channel_dropout_pca_development.csv.

SR6. Analyses not run (no results are reported for these)

The strict train-only scVI_matched/totalVI retraining on the high-confidence training partition is complete and is the primary internal analysis (Supplementary Methods S3; SR1b); it is not a blocked analysis. The following analyses were not run, and no result is claimed for them:

Table S12 | Analyses explicitly not run.

Analysis	Status
totalVI retraining under full marker-channel dropout	NOT RUN
Additional newly trained multimodal comparator (e.g., WNN, MultiVI, Cobolt)	NOT RUN (MOFA+ retained only as a historical secondary baseline)
Second external cohort with non-protein-gated labels	NOT RUN (no such cohort available locally)

Table S13 | Supplementary completeness checklist against main-text statements.

Main-text statement	Status
Strict train-only deep internal retraining (scVI_matched, totalVI)	COMPLETE — PRIMARY ANALYSIS
Class-stratified paired-bootstrap sensitivity	COMPLETE
Cell-type-specific Direction B degradation	COMPLETE
PCA-only marker-channel dropout	COMPLETE
Macro-F1 versus selective coverage	COMPLETE
Balanced accuracy versus selective coverage	COMPLETE
Lawlor donor median/SD/IQR/min/max and exact sign test	COMPLETE
Fixed-ontology Lawlor sensitivity	PARTIAL (column present; no separate divergent analysis)

Supplementary Data file index (05_GPB_Supplementary_Data.zip)

Table S14 | Files in the Supplementary Data archive.

File	Contents
strict_train_only/strict_train_only_point_estimates.csv	PRIMARY internal analysis. Held-out macro-F1 point estimates (RNA PCA, protein PCA, concat, scVI_matched, totalVI) and G_{access}, G_{assoc}, Gap_{total}, R_{access} under strict train-only fitting (7,573 train / 1,921 held-out high-confidence cells).
strict_train_only/strict_train_only_bootstrap_summary.csv	PRIMARY. Global cell-level conditional paired bootstrap summary (B = 10,000; seed 20260923): point estimates, bootstrap mean/median/SD, percentile 2.5/97.5 limits, and prop>0.
strict_train_only/strict_train_only_bootstrap_global_vs_stratified.csv	Supplementary Results SR1. Global (primary) versus class-stratified (sensitivity; seed 20260924) paired bootstrap under strict train-only fitting.
strict_train_only/run_meta.json	Run metadata for the strict train-only retraining (cell counts, gene/protein counts, scvi-tools and PyTorch versions, device, seed).
information_access/information_access_sensitivity_comparison.csv	Supplementary Results SR1b. Historical feature-transductive versus strict train-only (PRIMARY) summary. Historical G_{access} and Gap_{total} point values (+0.079, +0.113) in this file are differences of rounded F1 values; the frozen historical bootstrap output gives 0.0784 and 0.1124 (see historical_feature_transductive/bootstrap_global_vs_class_stratified.csv).
historical_feature_transductive/bootstrap_global_vs_class_stratified.csv	NOT PRIMARY. Historical feature-transductive internal bootstrap (deep models fit on all 10,849 cells' features), global and class-stratified. Retained only as the information-access sensitivity reported in SR1b.
degradation/class_degradation_dirB_focus_subsets.csv	Supplementary Results SR2. Direction B post-adaptation totalVI per-class F1 (mean over seeds) at clean versus protein corruption 0.85 for six T-cell focus subsets.
degradation/class_degradation_dirB_totalVI_p085_delta.csv	Supplementary Results SR2 (all classes). Same quantity for every cell type, with mean support.
selective_prediction/selective_prediction_dirB_totalVI_cov50_vs_100.csv	Supplementary Results SR3. Direction B totalVI at 50% versus 100% coverage (20 seeds): risk, accuracy, macro-F1, balanced accuracy.
selective_prediction/selective_prediction_macroF1_by_coverage.csv	Supplementary Results SR3. Accuracy, macro-F1, and balanced accuracy versus coverage for each direction and deep model (20 seeds).
lawlor/lawlor_all_methods_donor_level_summary.csv	Supplementary Results SR4. Donor-level (n = 10 biological donors) summary of present-class macro-F1 for all five representations.
lawlor/lawlor_all_methods_per_donor.csv	Supplementary Results SR4. Per-donor present-class macro-F1 for all five representations (deep models seed-averaged within donor).
lawlor/lawlor_donor_paired_deltas_present_class.csv	Supplementary Results SR4. Per-donor totalVI − scVI_matched paired differences.
lawlor/lawlor_donor_summary_stats.csv	Supplementary Results SR4. Paired-difference summary (mean, median, SD, IQR, min, max, donors favoring each model, exact two-sided sign test).
lawlor/lawlor_simple_baseline_fold_means.csv	Supplementary Results SR4 (continuity only). Historical fold-mean PCA aggregates; not the preferred donor-level aggregation used in main Figure 6.
marker_channel_dropout/marker_channel_dropout_pca_development.csv	Supplementary Results SR5. PCA-only marker-channel dropout on the development cohort; totalVI not retrained. RNA PCA in this file (0.665) is the fixed clean-data RNA PCA of this sensitivity pipeline and is not a reported result.
marker_channel_dropout/marker_channel_dropout_meta.json	Design metadata for SR5 (protein channels, scope, seed).
README.txt	Description of every file in the archive.